TT-E-543a
February 24, 1966

SUPERSEDING
Fed. Spec. TT-E-543
October 25, 1949

FEDERAL SPECIFICATION

ENAMEL, INTERIOR, UNDERCOAT, TINTS AND WHITE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of ready-mixed enamel for use as an undercoat for interior gloss and semigloss enamels.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-E-508—Enamel; Interior, Semigloss, Tints and White.
TT-P-56—Primer Coating (Primer-Sealer), Pigmented Oil, Plaster and Wallboard.
TT-P-143—Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-T-291—Thinner; Paint, Volatile Spirits (Petroleum-Spirits).
UU-W-101—Wallboard; Composition.

Federal Standards:

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595—Colors.

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REQUIREMENTS

3.1 **Pigment.** Any suitable pigments may be used provided the resulting product meets the requirements of this specification.

3.2 **Vehicle.** The vehicle shall consist of a long, straight-oil-modified phthalic alkyd resin of the air-drying type, together with suitable petroleum or terpene thinners and driers, which will yield a product conforming to the requirements of this specification.

3.3 **Quantitative requirements.** The enamel undercoat shall conform to the quantitative requirements specified in table I and 3.3.1.

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TABLE I. *Quantitative requirements of enamel undercoat*

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight of paint	52	—
Nonvolatile vehicle, percent by weight of vehicle	37	—
Phthalic anhydride, percent by weight of nonvolatile vehicle	23	—
Drying oil acids, percent by weight of nonvolatile vehicle	60	—
Water, percent by weight of paint	—	0.5
Coarse particles and skins (residue retained on No. 325 sieve), percent by weight of pigment	—	0.5
Consistency, Krebs-Stormer, shearing rate 200 r.p.m.:		
Grams	150	225
Equivalent K.U.	72	86
Drying time:		
Set-to-touch time, hours	1/2	2
Dry hard time, hours	—	7
Specular gloss (without correction for diffuse reflectance)	3	10
Daylight 45°, 0° directional reflectance (white paint)	82	—
Fineness of grind	4	—

3.3.1 *Dry opacity.* Not more than 6.0 milliliters (ml.) per square foot (630 square feet per gallon) of the white enamel undercoat (minimum apparent reflectivity 82 percent) shall be required to give a dry-film contrast ratio of 0.94. The minimum dry-film contrast ratio for tints applied to the same wet-film thickness in terms of apparent reflectivity shall be as specified in table II.

TABLE II. *Minimum dry-film contrast ratios for tints*

Apparent reflectivity of tint	Contrast ratio	Apparent reflectivity of tint	Contrast ratio
Percent		*Percent*	
80	0.95	70	0.97
78	.95	68	.98
76	.96	66	.98
74	.96	64	.98
72	.97	62	.99
		60 (and lower)	.99

3.4 Qualitative requirements.

3.4.1 *Condition in container.* The enamel undercoat shall be free from grit, seediness, skinning, or livering in a freshly opened full container and shall show no more pigment settling or caking than can be readily redispersed with a paddle to a homogeneous state.

3.4.2 *Color.* The enamel undercoat shall be of the color specified (see 6.2) and shall match the appropriate color chip of Fed. Std. No. 595.

3.4.3 *Flexibility.* The enamel undercoat shall show no evidence of cracking when subjected to the flexibility test.

3.4.4 *Rosin and rosin derivatives.* Rosin and rosin derivatives shall not be present.

3.4.5 *Skinning.* A three-quarters filled closed container of the enamel undercoat, after 48 hours' storage at room temperature (70° to 90°F.), shall show no skinning.

3.4.6 *Brushing, flowing, spreading, and leveling properties.* The enamel undercoat shall show easy brushing and good flowing, spreading, and leveling properties.

3.4.7 *Appearance of coated surface.* The enamel undercoat shall dry to a uniform, smooth, flat appearance. There shall be no shiners, flashes, laps, or brush marks. Tinted enamel undercoats shall dry to a uniform color.

3.4.8 *Odor.* The odor of the enamel undercoat in the container and during and after application shall not be obnoxious.

3.4.9 *Reducibility with mineral spirits.* The enamel undercoat shall be compatible with mineral spirits conforming to grade 1 of TT-T-291, upon dilution of one volume of the package material as received with one volume of mineral spirits. The reduced product shall remain stable and homogeneous (except for settling of pigments) for a minimum of 4 hours.

3.4.10 *Enamel holdout.* The specular gloss of a semigloss enamel conforming to

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TT-E-508 applied over the enamel undercoat as specified in 4.4.2 shall not decrease more than 15. The appearance of the laminated composition wallboard panel, prepared and recoated with enamel conforming to TT-E-508 as specified in 4.4.1, shall also be satisfactory in regard to holdout.

3.4.11 *Storage stability.* A full, closed container of the enamel undercoat upon being opened 1 year after date of manufacture, when stored under warehouse conditions (70° to 90°F.), shall show no livering, skinning, curdling, or gelling and no more settling than can be readily redispersed with a paddle to a homogeneous state. The viscosity of the stored enamel undercoat shall not exceed 96 Krebs units and the dry hard time shall not exceed 7 hours.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).

(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 *Sampling and inspection.* Sampling and inspection shall be in accordance with methods 1011, 1021, and 1031.1 of Fed. Test Method Std. No. 141, as applicable.

4.4 Test procedures. The enamel undercoat shall be tested in accordance with the following applicable test methods of Fed. Test Method Std. No. 141 indicated in table III, and as specified hereinafter. Failure of any test shall be cause for rejection of the lot which the sample represents.

TABLE III. *Tests and methods*

Examination or test	Applicable method in Fed. Std. No. 141	Requirements paragraphs
Condition in container	3011	3.4.1
Pigment content	4021	Table I
Nonvolatile matter in vehicle	4052	Table I
Drying time	4061	Table I
Water	4081	Table I
Coarse particles and skins	4091	Table I
Dry opacity	4121	3.3.1
Storage stability	4142	3.4.11
Skinning	4141	3.4.5
Color	[1]4250	3.4.2
Consistency	4281	Table I
Reducibility and dilution stability	4203	3.4.9
Odor	4401	3.4.8
Fineness of grind	4411	Table I
Rosin and rosin derivatives	5031	3.4.4
Specular gloss	6101	Table I
Directional reflectance	6121	Table I
Flexibility	[2]6221	3.4.3
Phthalic anhydride	7021	Table I
Oil acids	7031	Table I

[1] Apply the film to a clean, smooth, plate-glass panel with a 0.003-inch (approximately 0.006-inch gap clearance) Bird film applicator, or any other doctor blade which produces a film of the same thickness as that produced by the Bird blade, and allow to dry for 24 hours.

[2] Apply a film of the paint to a flat tin panel (method 2013) using a 0.002-inch (approximately 0.004-inch gap clearance) Bird film applicator or a similar doctor blade. Air dry for 18 hours, bake 5 hours at 105° ± 2°C., cool to 23° ± 1°C., bend double rapidly over a 1/2-inch mandrel, and examine.

4.4.1 *Brushing, flowing, spreading, and leveling properties; appearance of dried film.* Prepare a panel of laminate composition wallboard conforming to UU-W-101, at least 8 square feet in area, by ap-

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plying two uniform brush coats of plaster and wallboard primer-sealer conforming to TT-P-56 to the smooth side. Allow the first coat to dry for 24 hours before the second is applied and allow the latter also to dry for 24 hours. Place the panel in a vertical position and apply the well-mixed enamel undercoat with a 4-inch wall brush at a spreading rate of about 600 square feet per gallon. In applying the undercoat, first brush across a section of the panel with back-and-forth strokes, then level the coat with light vertical strokes, using the tip of the brush. Proceed to the next section in the same manner, always working to a "wet edge". Note whether the undercoat brushes easily and if it has good flowing and spreading properties. Let the panel dry for 24 hours. Observe whether the film has flowed out to a level, uniform appearance. Inspect for freedom from shiners, laps, and brush marks. Re-coat uniformly with semigloss enamel conforming to TT-E-508 at a spreading rate of about 500 square feet per gallon using a 2-1/2-inch brush. Inspect after 24 hours drying for nonuniformity of gloss or other evidence of inferior enamel holdout. Nonconformance to 3.4.6, 3.4.7, and 3.4.10 shall constitute failure of this test.

4.4.2 *Enamel holdout.* Apply a film of the enamel undercoat to a clean, smooth, plate-glass panel with a 0.0025-inch (approximately 0.005-inch gap clearance) Bird film applicator, or similar doctor blade, and allow to dry for 48 hours at a temperature of 23° ± 1.1°C. and a relative humidity of 50 ± 4 percent. Then apply (crosswise) to the undercoat and to a separate, clean, plate-glass panel a coat of semigloss enamel conforming to TT-E-508, using a 0.0015-inch (approximately 0.003-inch gap clearance) Bird film applicator, or similar doctor blade. Let both panels dry for 48 hours at the same temperature and humidity as above. Measure the specular gloss of both panels and determine the difference. Nonconformance to 3.4.10 shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested in accordance with TT-P-143 to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The enamel undercoat shall be packaged, packed, and marked in accordance with TT-P-143 with the following exception: Wood cleated fiberboard, wood cleated paper overlaid, and fiber boxes are not acceptable for level A packing. Preservation and packaging shall be level A, B, or C and packing shall be A, B, or C as specified (see 6.2). The enamel undercoat shall be packaged in the size of container specified (see 6.2).

5.1.1 *Additional marking.* In addition to the marking requirements specified in 5.1, each container shall be marked with the following instructions:

"New dry-plaster surfaces and composition wallboard should first be sealed properly with one or two coats, as required, of primer sealer (TT-P-56). Interior masonry and brick should be dry, and usually requires two coats of a special alkali resistant primer, if portland cement is present, to prepare the surface for the enamel undercoat. Old, glossy, painted surfaces which are in good condition and not dirty may be coated directly with the undercoat before applying a new coat of gloss enamel. New wood may be primed with the undercoat thinned with 1 pint of mineral spirits or turpentine per gallon of undercoat, and this followed by one coat of the undercoat as received. The undercoat may be lightly sanded, after it has dried thoroughly, before applying a gloss enamel."

6. NOTES

6.1 Intended use. The enamel undercoat covered by this specification is intended specifically for use as an undercoat for interior gloss and semigloss enamels. It is

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characterized by easy brushing, rapid drying, smooth leveling, and high opacity.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Color required (see 3.4.2).

(*c*) Levels of packaging and packing required (see 5.1). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the enamel will be held in covered storage approximately 1 year from date of initial preservation and packaging.

(*d*) Size of container required (see 5.1).

6.3 **Basis of purchase.** The enamel undercoat covered by this specification should be purchased by volume, the unit being one U. S. gallon of 231 cubic inches at 15.5°C. (60°F.).

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 **Transportation descriptions.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Paints, not otherwise indexed by name, liquid.
Carload minimum weight 36,000 pounds.

Motor:

Paints, not otherwise indexed, liquid.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

GSA
Army—MO(ERDL)
Navy—SH
Air Force—69

Review activities:
Army—MO, MD, CE
Navy—SH
Air Force—69
DOD Misc—NSA

User activities:
Army—EL, GL, MI, MR
Navy—CG, MC
Air Force—None

Civil interest:
AGR
COM
DC
GSA
PO
VA

Preparing activity:
Army—MO(ERDL)

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